

## Coordinates

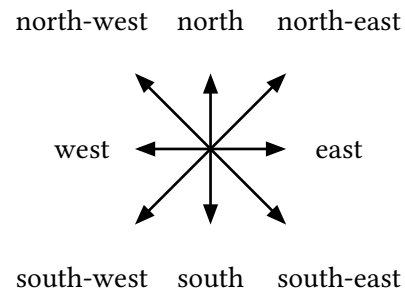
This package adopts the CeTZ coordinate system. All of CeTZ's coordinate definitions also work within this package. For further reading consult the CeTZ documentation.

To simplify the process of drawing track infrastructure, several options for defining coordinates have been added.

### Relative position in cardinal direction

Support for drawing tracks in all eight directions.

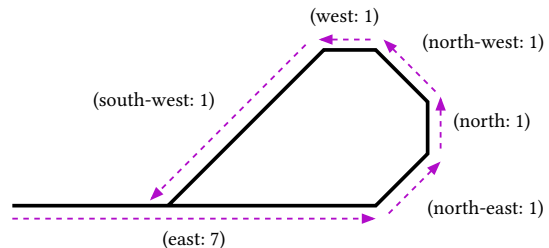
For example (e: 2) describes to go two units east and is internally translated to: (rel: (2, 0))



( key : number )

key: one of the cardinal directions: north, north-east, east, south-east, south, south-west, west, north-west.

```
track(  
  (),  
  (east: 7),  
  (north-east: 1),  
  (north: 1),  
  (north-west: 1),  
  (west: 1),  
  (south-west: 3)  
)
```



### Position on track

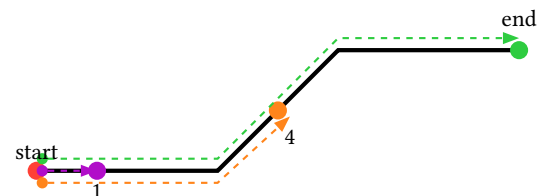
To position elements on a referenced track.

str . number or (name: str , anchor: number )

The syntax is identical to the anchor syntax of CeTZ. For tracks the default functionality of CeTZ is overwritten to match with the length calculations of track schematics.

The default anchors start and end are also supported.

```
track(  
  (), (east: 3), (north-east: 2), (east: 3),  
  name: "tr-1"  
)  
circle("tr-1.start", fill: red)  
circle("tr-1.1", fill: purple)  
// long syntax:  
circle(  
  (name: "tr-1", anchor: 4),  
  fill: orange  
)  
circle("tr-1.end", fill: green)
```



## Track intersection in cardinal direction

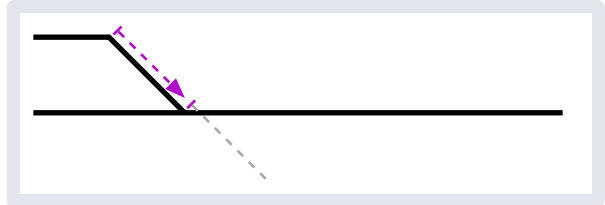
Draw the track until it intersects with the specified track.

( key : str )

key: one of the cardinal directions: north, north-east, east, south-east, south, south-west, west, north-west.

str: track name.

```
track((), (east: 7), name: "tr-1")
track((0, 1), (east: 1), (south-east:
"tr-1"))
```



## Elements

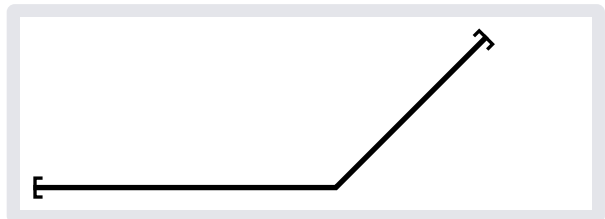
- buffer-stop()
- track()
- turnout()

### buffer-stop

Draws a buffer stop

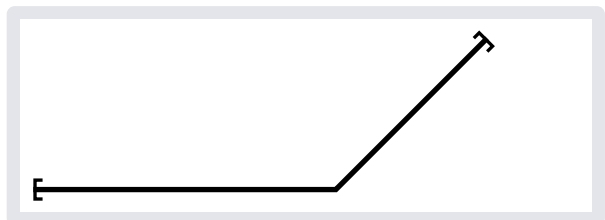
The direction is inferred from the track. Buffer stops must be drawn at the start or the end of a track.

```
track((), (east: 4), (north-east: 2),
name: "tr-1")
buffer-stop(())
buffer-stop("tr-1.start")
```



Buffer stops can also be added directly to a track with the start and end parameter.

```
track(
    (), (east: 4), (north-east: 2),
    start: buffer-stop,
    end: buffer-stop,
    name: "tr-1"
)
```



## Parameters

```
buffer-stop(
    point: coordinate,
    stroke: auto none length color gradient stroke tiling dictionary
)
```

**point** coordinate

Positional coordinate to draw the buffer stop.

**stroke** `auto` or `none` or `length` or `color` or `gradient` or `stroke` or `tiling` or dictionary

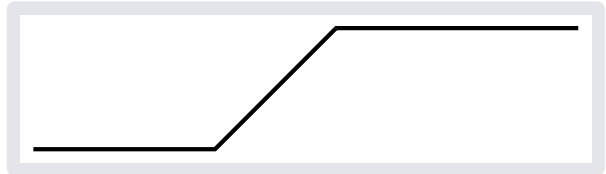
How to stroke the buffer stop. See Typst's line documentation for more details.

Default: `auto`

## track

Draws a track

```
track(), (east: 3), (north-east: 2),  
(east: 4))
```



### Parameters

```
track(  
  ..points: coordinate,  
  kind: str,  
  stroke: none length color gradient stroke tiling dictionary,  
  start: none function,  
  end: none function,  
  name: none str  
)
```

**..points** `coordinate`

Positional two or more coordinates to draw tracks.

**kind** `str`

Kind of track. Can be either "main" or "secondary".

```
track(), (east: 7))
```



```
track(), (east: 7), kind: "secondary")
```



Default: `"main"`

**stroke** `none` or `length` or `color` or `gradient` or `stroke` or `tiling` or dictionary

How to stroke the track. See Typst's line documentation for more details.

Default: `auto`

**start** none or function

Element at start of track e.g. buffer-stop

```
track(  
  (), (east: 2),  
  start: buffer-stop,  
  name: "tr-1"  
)
```



Elements at the end of track are aliased:

- "]" → buffer-stop

```
track(  
  (), (east: 2),  
  start: "]",  
  name: "tr-1"  
)
```



Default: none

**end** none or function

Element at end of track e.g. buffer-stop

```
track(  
  (), (east: 2),  
  end: buffer-stop,  
  name: "tr-1"  
)
```



Aliases can also be used. See start attribute.

Default: none

**name** none or str

Anchor name.

Default: none

## turnout

Draws a turnout.

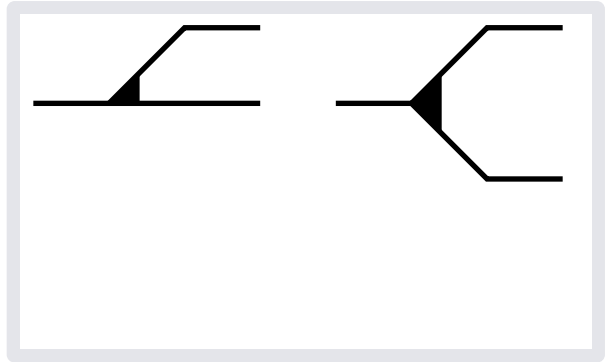
Connecting tracks are automatically detected and a turnout is drawn as soon as the turnout is connected on all three sides. The Symbol is automatically placed in the correct orientation.

Turnouts are only drawn if they match the angle pattern of the two types:

- straight + 45° diverging branch
- two 45° diverging branches

```
track((, (east: 3), name: "tr-1")
turnout("tr-1.1")
track((, (north-east: 1), (east: 1))

track(
  (4,0), (east:1), (north-east:1),
  (east:1),
  name: "tr-2"
)
turnout("tr-2.1")
track((, (south-east: 1), (east:1))
```



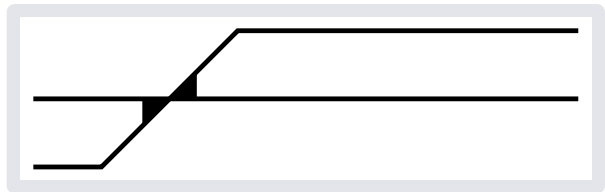
Three way switches are supported

```
track((, (east: 4), name: "tr-1")
turnout("tr-1.1", name: "to-1")
track("to-1", (south-east: 1), (east: 2))
track("to-1", (north-east: 1), (east: 2))
```



With four connections the switch converts to a slip switch

```
track((, (east: 8), name: "tr-1")
track((0,-1), (east:1), (north-east: 2),
  (east: 5))
turnout("tr-1.2")
```



## Parameters

```
turnout(
  point: coordinate,
  fill: auto none color gradient tiling,
  stroke: auto none length color gradient stroke tiling dictionary,
  name: none str
)
```

**point** coordinate

Positional coordinate to draw the turnout.

**fill** auto or none or color or gradient or tiling

How to fill the turnout.

Default: auto

**stroke** auto or none or length or color or gradient or stroke or tiling or dictionary

How to stroke the turnout. See Typst's line documentation for more details.

Default: auto

**name** `none` or `str`

Anchor name.

Default: `none`